

MF³: Efficient Multimodal Federated Learning with Dual-Path Mamba-Transformer for Metro Flow Prediction

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Abstract

Metro flow prediction is a critical application in smart city and Web of Things infrastructures, essential for optimizing urban mobility. However, building such predictive systems faces three key challenges: (1) the fragmentation of multimodal spatiotemporal data, (2) the inefficiency of existing models in capturing long-range dependencies, and (3) the data silos and privacy concerns inherent in distributed station infrastructures. To address these challenges, a multimodal federated learning framework named MF³ (Mamba-Transformer-Federated Metro Flow Prediction) is proposed. First, a multimodal alignment (MA) module is designed, where cross-modal alignment attention bridges visual and spatiotemporal features, thereby enhancing feature complementarity and alignment. Second, a dual-path Mamba-Transformer (DMT) module is designed, in which Mamba's linear long-range memory and the Transformer's global perception operate in parallel, reducing information loss. Third, a blockchain-based federated reputation (BFR) module is established to perform personalized federated learning, thereby enhancing privacy protection. Finally, extensive experiments on real metro datasets from Hangzhou and Shanghai demonstrate that MF³ achieves superior performance in terms of prediction accuracy. In summary, the proposed MF³ framework provides a new feasible paradigm for metro flow prediction, supporting urban traffic optimization, metro operation and scheduling, and the development of smart city and Web of Things infrastructures.

CCS Concepts

• **Computing methodologies** → **Machine learning**: *Distributed computing methodologies*; • **Information systems** → **Spatiotemporal systems**.

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Keywords

Metro flow prediction; Multimodal alignment; Dual-path Mamba-Transformer; Federated learning

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1 Introduction

Urban rail transit systems, especially metro networks, are a critical component of modern metropolitan infrastructures and play an essential role in Web of Things (WoT)-enabled smart city ecosystems [25, 42]. Unlike traditional traffic flow prediction that primarily deals with road networks characterized by irregular topology and vehicle flow, metro flow prediction focuses on structured, station-based passenger movements constrained by fixed routes and schedules, making it inherently multimodal and spatiotemporally dependent. With the widespread deployment of sensing devices, mobile applications, and automated fare collection systems, massive amounts of multimodal passenger flow data are continuously generated and connected through the web-based infrastructure of urban mobility services [23, 26]. These data streams provide unprecedented opportunities to enhance operational efficiency, support real-time passenger services, and enable the seamless integration of metro systems into broader WoT infrastructures [22]. Consequently, accurate metro passenger flow prediction is not only a core task in intelligent transportation management but also a fundamental requirement for building reliable systems and infrastructures that optimize train dispatch intervals, reduce passenger waiting times, and alleviate congestion in WoT-driven urban environments [14, 38, 41]. Nevertheless, reliable prediction remains challenging due to multimodal and fragmented data distributions, long-term temporal dependencies coupled with local dynamics, and decentralized station-level data storage that complicates both privacy protection and federated collaboration.

Early studies primarily rely on statistical methods, such as regression analysis [9] and time-series modeling [31], to perform metro flow prediction. Although these approaches are simple and interpretable, they often fail to capture nonlinear dynamics and complex spatial dependencies inherent in metro systems. To overcome these

limitations, machine learning techniques such as decision trees and support vector machines are gradually introduced, offering stronger predictive power [37]. However, these models still lack the ability to effectively represent sequential correlations across time. With the rise of deep learning, recurrent neural networks (RNNs) [11], long short-term memory (LSTM) networks [27], and gated recurrent units (GRUs) [40] become dominant, as they are able to capture temporal dependencies more comprehensively. Nevertheless, their sequential nature limits scalability and makes it difficult to model long-range dependencies. Furthermore, to address spatial complexity, graph neural networks (GNNs) [7, 13] are proposed to handle non-Euclidean structures and uncover inter-station relationships. However, their effectiveness often diminishes when dealing with multimodal and heterogeneous inputs. More recently, Transformers demonstrate strong capabilities in modeling long-range temporal dependencies through attention mechanisms [14], while the emerging Mamba model opens new opportunities for efficient long-sequence processing with linear complexity and enhanced local feature extraction [10, 15]. Despite these advancements, existing approaches still fall short in WoT-driven urban infrastructures, where metro flow prediction must be accurate, robust to multimodal inputs, and efficient under decentralized and privacy-sensitive environments. Consequently, these gaps call for new frameworks that can not only enhance predictive performance but also align with the requirements of Web, mobile, and WoT infrastructures, thereby motivating the design of the proposed MF³ framework.

Building upon the recent developments and their limitations, it becomes evident that existing models still struggle to meet the demands of WoT-integrated metro systems. Specifically, metro passenger flow prediction in WoT-enabled infrastructures faces three key challenges:

(1) Heterogeneous modalities pose a significant challenge to effective fusion. Multimodal data from time series, spatial topologies, and external factors are heterogeneous and fragmented, making cross-modal fusion in WoT systems difficult.

(2) Global attention mechanisms suppress the modeling of local dynamics. Transformers struggle with long-sequence efficiency, where global attention often suppresses local dynamics, limiting accurate modeling of both short- and long-term patterns.

(3) Decentralized data silos hinder effective collaborative modeling. Metro stations operate as decentralized data silos, and in WoT infrastructures, privacy protection and communication overhead hinder effective collaborative modeling.

To address these challenges, a privacy-preserving multimodal federated learning (FL) framework, MF³ (Multimodal FL with Dual-path Mamba-Transformer), is proposed. MF³ consists of three core modules: MA (Multimodal Alignment), DMT (Dual-Path Mamba-Transformer), and BFR (Blockchain-based Federated Reputation), which correspond to the key functional components of the framework. Specifically, MA enhances cross-modal feature fusion, DMT improves long-sequence pattern capture, and BFR enables privacy-aware collaborative modeling across decentralized stations. Figure 1 provides an overview of the proposed MF³ framework, illustrating the integration of metro flow prediction within the selected WoT-driven infrastructure and highlighting the interactions among its three major modules. By synergizing these components, the framework enables efficient collaborative training while strictly

adhering to privacy constraints across the distributed network.

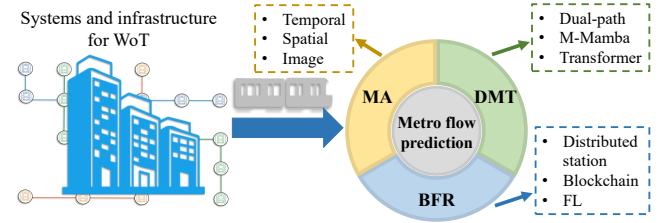


Figure 1: The overall architecture of the proposed MF³ framework. It supports the construction of reliable systems and infrastructures for Web, mobile, and WoT environments through metro flow prediction.

First, the MA module is designed to mitigate the heterogeneity across modalities. Specifically, it generates heatmaps representing weekday/weekend in/outflows for each station. By incorporating data from neighboring stations, this visual modality intuitively captures dynamic flow distributions. Consequently, imaging-based encoding enhances cross-modal fusion robustness, laying a solid foundation for accurate prediction in WoT infrastructures.

Second, the DMT module with a fusion gate is designed to capture both local fluctuations and long-term dependencies. In the temporal path, a dual-channel embedding combines fine-grained time series with periodic features. The spatial path leverages heatmaps and adjacency information to jointly learn spatial-visual dependencies. This efficient long-sequence modeling ensures predictive robustness within smart transportation networks.

Third, to address privacy and communication challenges in decentralized metro systems, the BFR module is designed. The framework employs reputation-based aggregation to prioritize reliable clients and uses similarity clustering for personalized model delivery. This approach significantly reduces communication overhead while enhancing fairness and robustness for privacy-strict WoT environments.

The main contributions of this paper can be summarized as follows:

(1) MA for unified metro flow prediction. An effective multimodal embedding and alignment mechanism is designed to integrate temporal, spatial, and visual modalities. In particular, self-generated metro flow heatmaps are introduced as visual features, which directly reflect passenger dynamics and enhance cross-modal fusion robustness.

(2) DMT for efficient spatiotemporal modeling. The MambaFormer model is proposed, incorporating a dual-path fusion gate that combines Mamba's linear-complexity local feature extraction with Transformer's global dependency modeling. This design captures complex spatiotemporal dependencies efficiently and effectively.

(3) BFR for robust and fair federated aggregation. A dynamic reputation mechanism built on blockchain is developed to support adaptive model updates. This approach significantly reduces communication overhead, improves system robustness against unreliable participants, and strengthens privacy protection in distributed metro environments.

(4) Comprehensive empirical validation. Extensive experiments are conducted on real-world metro datasets. The results show that MF³ achieves significant performance improvements compared with the strongest available baselines, with up to 55.10% improvement for MAE, 38.29% for MAPE, and 47.32% for RMSE on the Shanghai dataset, demonstrating its superior accuracy, efficiency, and privacy preservation.

Through these contributions, MF³ offers a scalable and secure framework for metro passenger flow prediction, effectively integrating multimodal learning, efficient long-sequence modeling, and trustworthy FL, while addressing the challenges of decentralized data and privacy constraints inherent in WoT-enabled urban infrastructures.

2 Problem and Method

In this section, the metro flow prediction problem is introduced by clarifying the basic terminology and providing a formal definition. Based on this formulation, the proposed MF³ framework is then presented, followed by a detailed explanation of its methodological components. Finally, the overall architecture of MF³ is illustrated in Figure 2, highlighting how the framework supports accurate and scalable metro flow prediction within WoT-enabled urban infrastructures and distributed smart transportation systems.

2.1 Problem Statement

Before delving into the details of the MF³ framework, it is helpful to provide an overview of the metro flow prediction problem and its key concepts, facilitating a clearer understanding of the task addressed by the model.

Let the metro system be represented by a directed graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}), \quad (1)$$

where each vertex $v_i \in \mathcal{V}$ is a station and each edge $(v_i, v_j) \in \mathcal{E}$ is a physical track segment.

The metro flow data are represented using three complementary components:

- Time-series tensor:

$$\mathbf{X} = \{X_i^{(t)} \in \mathbb{R}^d \mid v_i \in \mathcal{V}, t \in \mathcal{T}\}, \quad (2)$$

where $X_i^{(t)}$ stacks d -dimensional measurements, including entries, exits, and transfers, for the station v_i in the time slot t . This tensor captures the temporal dynamics of passenger flows across all stations.

- Spatial adjacency:

$$\mathbf{A} \in \{0, 1\}^{|\mathcal{V}| \times |\mathcal{V}|}, \quad (3)$$

where $\mathbf{A}$ represents the connectivity of the metro network. Specifically, $A_{ij} = 1$ if the stations v_i and v_j are directly connected, and $A_{ij} = 0$ otherwise. This adjacency matrix encodes the spatial topology of the rail network.

- Flow-density heatmap:

$$\mathbf{H}_i^{(t)} \in \mathbb{R}^{h \times w}, \quad (4)$$

where $\mathbf{H}_i^{(t)}$ represents pixel-level passenger density on the platform of the station v_i at time t , capturing intra-station spatial distributions.

Given the historical observations $\mathcal{D}_{\text{hist}} = \{\mathbf{X}, \mathbf{A}, \mathbf{H}\}$ up to the current time-stamp T , the goal is to learn a predictive mapping:

$$\mathcal{F}_\theta : (\mathbf{X}^{(T-\tau+1:T)}, \mathbf{A}, \mathbf{H}^{(T-\tau+1:T)}) \longrightarrow \hat{\mathbf{Y}}^{(T+1:T+p)}, \quad (5)$$

where $\hat{\mathbf{Y}}^{(T+1:T+p)}$ denotes the predicted passenger flows for each station v_i over the future horizon $k = T + 1, \dots, T + p$. This formulation unifies temporal, spatial, and visual information to support accurate multi-step flow prediction.

The resulting predicts are injected into the automatic train-regulation engine to optimize headways Δt_{head} in real time, minimise the average platform dwell time $\bar{w}_{\text{plat}}$, and generate early crowd-guidance alerts.

2.2 Data Preprocessing

Before model training, the raw metro data must be preprocessed to ensure consistency across modalities. Proper preprocessing helps reduce noise, align heterogeneous data sources, and enhance feature representation quality, thereby laying a solid foundation for multimodal fusion.

Flow data normalization. To address the long-tailed distribution of metro flow data and mitigate the impact of extreme values, a logarithmic transformation followed by min-max normalisation is applied to the inflow and outflow time series. Specifically, for each station i and time step t , the raw flow value $x_i(t)$ is transformed as:

$$x_{\text{norm}} = \frac{\log(1 + \max(0, x_i(t)))}{\max_{x \in X} \log(1 + x)}, \quad (6)$$

where X represents all flow values in the training set. This approach effectively handles the sparse nature of metro flow data, particularly during off-peak hours.

Temporal feature engineering. Beyond the flow values themselves, rich temporal features are extracted to capture periodic patterns in metro usage:

- Circular time encoding: Hour of day is encoded using sine and cosine transformations to preserve cyclical continuity:

$$\text{hour_sin} = \sin\left(\frac{2\pi \cdot \text{hour}}{24}\right), \quad \text{hour_cos} = \cos\left(\frac{2\pi \cdot \text{hour}}{24}\right). \quad (7)$$

- Day type indicators: Binary flags for weekends/holidays and peak hours (7-9 AM, 5-7 PM)
- Sequential temporal context: The immediate historical flow patterns are incorporated through the station's ego-graph neighborhood

Self-generated heat-maps. To provide spatial-visual context of crowd dynamics around each station, four daily heat-maps are generated per station. Formally, for the station v_i with its neighborhood $\mathcal{N}(i) = \{j \mid (v_i, v_j) \in \mathcal{E}\} \cup \{i\}$, flow patterns are aggregated by:

$$\mathbf{H}_i^{d,f} \in \mathbb{R}^{96 \times |\mathcal{N}(i)|}, \quad \mathbf{H}_i^{d,f}(t, n) = \frac{1}{|D_d|} \sum_{\delta \in D_d} x_n^f(t, \delta), \quad (8)$$

where $t = 0, 1, \dots, 95$ indexes 15-minute intervals, $n \in \mathcal{N}(i)$, D_d contains all dates of type $d \in \{\text{weekday}, \text{weekend}\}$, and $x_n^f(t, \delta)$ is the raw flow. Each heat-map row is min-max normalized to $[0, 1]$ and converted to RGB images, providing visual priors for the multimodal learning framework.

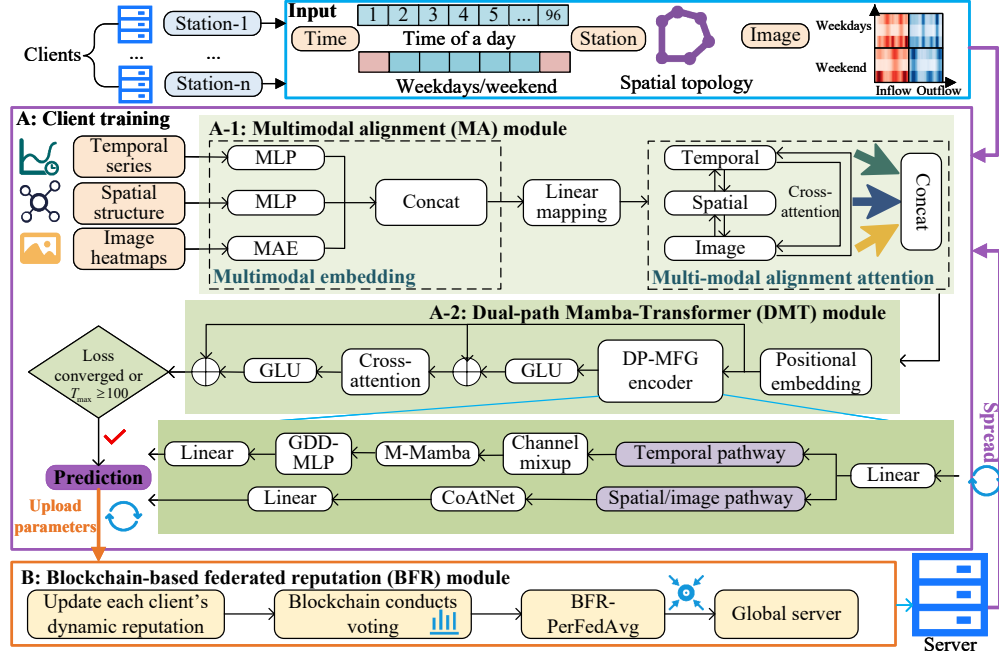


Figure 2: The Workflow of MF³ for metro flow prediction. It consists of three modules: MA for multimodal alignment, DMT for efficient spatiotemporal modeling, and BFR for privacy-preserving federated aggregation.

2.3 Multimodal Alignment

To fully exploit the complementary information from heterogeneous data sources, it is crucial to align their representations before fusion. In metro systems, temporal sequences, spatial structures, and visual heatmaps differ in characteristics and scale, making direct integration suboptimal. To address this, the MA module is constructed, which aligns these modalities into a unified latent space and facilitates effective cross-modal feature interaction.

Specifically, the MA module transforms each modality into modality-specific embeddings before alignment. Temporal sequences $X_t \in \mathbb{R}^{T \times d_t}$ and spatial structures $X_s \in \mathbb{R}^{S \times d_s}$ are first projected into a unified latent space through dedicated multi-layer perceptrons (MLPs):

$$\begin{aligned} Z_t &= \text{MLP}_{\text{temporal}}(X_t) = \text{ReLU}(X_t W_t^{(1)} + b_t^{(1)}) W_t^{(2)} + b_t^{(2)}, \\ Z_s &= \text{MLP}_{\text{spatial}}(X_s) = \text{ReLU}(X_s W_s^{(1)} + b_s^{(1)}) W_s^{(2)} + b_s^{(2)}. \end{aligned} \quad (9)$$

Concurrently, station heatmap images $X_i \in \mathbb{R}^{H \times W \times 3}$ are processed through a pre-trained Masked Autoencoder (MAE) to extract rich visual representations:

$$Z_i = \text{MAE}_{\text{visual}}(X_i). \quad (10)$$

The three modality embeddings are subsequently concatenated and linearly projected to ensure dimensional consistency:

$$Z_{\text{concat}} = \text{Linear}([Z_t \| Z_s \| Z_i]), \quad (11)$$

where $\parallel$ denotes the concatenation operation and the linear projection adjusts the combined dimensionality to d_{hidden} .

The core innovation of the MA module lies in its comprehensive cross-modal attention mechanism. For each of the six possible directional pairs among the three modalities, bidirectional cross-attention is performed:

$$\begin{aligned} \text{CrossAttn}_{m \rightarrow n} &= \text{Softmax} \left(\frac{(Z_m W_m^Q)(Z_n W_n^K)^T}{\sqrt{d_k}} \right) (Z_n W_n^V), \\ \text{CrossAttn}_{n \rightarrow m} &= \text{Softmax} \left(\frac{(Z_n W_n^Q)(Z_m W_m^K)^T}{\sqrt{d_k}} \right) (Z_m W_m^V), \end{aligned} \quad (12)$$

where $m, n \in \{\text{temporal, spatial, visual}\}$ represent distinct modalities. This results in six cross-attention outputs that capture all pairwise modal interactions.

Finally, these cross-modal representations are concatenated to form the comprehensive multimodal embedding:

$$Z_{\text{ma}} = \text{Concat} \left(\text{CrossAttn}_{t \rightarrow s}, \text{CrossAttn}_{s \rightarrow t}, \text{CrossAttn}_{t \rightarrow i}, \text{CrossAttn}_{i \rightarrow t}, \text{CrossAttn}_{s \rightarrow i}, \text{CrossAttn}_{i \rightarrow s} \right). \quad (13)$$

This meticulously designed alignment strategy ensures that temporal patterns are contextualized with spatial constraints and visual crowd dynamics, while spatial configurations are informed by temporal evolution and visual context. The output Z_{ma} serves as the

input to the subsequent DMT module, carrying rich, cross-modally aligned representations for further processing.

2.4 Dual-Path Mamba-Transformer

The DMT module represents a sophisticated fusion of Mamba's selective state space mechanisms with Transformer's attention capabilities, designed to capture multi-scale spatiotemporal dependencies in metro flow data. The processing pipeline begins with positional encoding of the multimodal-aligned input:

$$Z_{in} = Z_{ma} + PE(t), \quad (14)$$

where $PE(t)$ denotes the sinusoidal positional embeddings that preserve temporal ordering information.

The encoded input is then processed by the DMT encoder, which operates through two specialized pathways. The temporal pathway is further divided into dual channels to capture complementary temporal dynamics:

$$\begin{aligned} Z_{fine} &= MLP_{fine}(Z_{in}), \\ Z_{periodic} &= MLP_{periodic}(Z_{in}). \end{aligned} \quad (15)$$

The fine-grained channel processes high-resolution temporal data to model continuous passenger flow variations, while the periodic channel focuses on rhythmic patterns based on weekly cycles. Both temporal representations are subsequently integrated and processed through the M-Mamba module [34], a modified variant of Mamba proposed in CMamba [34], which enhances the original state space model by introducing data-dependent mechanisms and improved cross-time dependency modeling tailored for multivariate time series prediction:

$$Z_{temporal} = \text{M-Mamba}(\text{Concat}[Z_{fine}, Z_{periodic}]). \quad (16)$$

Followed by gated dilated deep MLP (GDD-MLP) for multi-scale feature extraction:

$$Z_{temp_out} = \text{Linear}(\text{GDD-MLP}(Z_{temporal})). \quad (17)$$

Concurrently, the spatial/image pathway processes self-generated heatmaps alongside station spatial adjacency information to model spatiotemporal dependencies at both local and global scales. This pathway employs a convolutional attention network (CoAtNet) for joint representation learning:

$$Z_{spatial} = \text{Linear}(\text{CoAtNet}(Z_{in})). \quad (18)$$

Upon exiting the DP-MFG encoder, outputs from both pathways undergo gated fusion. First, a gated linear unit (GLU) is applied to the combined representations:

$$Z_{glu} = \text{GLU}(\text{Concat}[Z_{temp_out}, Z_{spatial}]). \quad (19)$$

Followed by residual connection with the originally position-encoded input:

$$Z_{res1} = Z_{glu} + Z_{in}. \quad (20)$$

The refined representations then proceed through cross-attention mechanisms that facilitate interaction between temporal and spatial modalities:

$$\begin{aligned} Q &= Z_{res1} W^Q, \quad K = Z_{res1} W^K, \quad V = Z_{res1} W^V, \\ Z_{attn} &= \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right) V. \end{aligned} \quad (21)$$

Another GLU operation is applied to enhance feature selectivity:

$$Z_{glu2} = \text{GLU}(Z_{attn}). \quad (22)$$

Followed by a final residual connection with the position-encoded input:

$$Z_{out} = Z_{glu2} + Z_{in}. \quad (23)$$

This meticulously designed architecture ensures comprehensive modeling of both transient flow variations and persistent spatial constraints, while the hierarchical fusion strategy enables effective information exchange across temporal and spatial domains. The output Z_{out} carries rich spatiotemporal representations suitable for final flow prediction tasks.

2.5 Blockchain-Based Federated Reputation

To address the challenges of client data heterogeneity and model quality variations in FL environments, the BFR module is designed. This mechanism enhances global model performance while preserving data privacy through distributed reputation assessment and weighted aggregation strategies.

Client Model Upload and Reputation Initialization. Each client c_i uploads a triple to the server after local training:

$$\mathcal{U}_i = \langle i, \Theta_i, \mathcal{L}_i^{\text{val}} \rangle, \quad (24)$$

where Θ_i represents the model parameters of the client i , and $\mathcal{L}_i^{\text{val}}$ denotes the validation loss. The federated system initializes the reputation book:

$$\mathcal{R}^0 = \{r_i^0 = r_{\text{init}} \mid \forall i \in [N]\}, \quad (25)$$

where r_{init} is the initial reputation value, and N is the total number of clients.

Dynamic reputation update mechanism. Based on client-reported validation performance, reputation scores are updated via exponential moving average:

$$r_i^{t+1} = \alpha_{\text{ema}} \cdot r_i^t + (1 - \alpha_{\text{ema}}) \cdot \phi(\mathcal{L}_i^{\text{val}}), \quad (26)$$

where $\alpha_{\text{ema}} \in [0, 1]$ is a smoothing factor, and the transformation function $\phi(\mathcal{L}) = \frac{1}{1+\mathcal{L}}$ maps losses to reputation scores in the $[0, 1]$ interval.

Multi-dimensional voting weight calculation. The improved blockchain voting mechanism computes aggregation weights by integrating three dimensions: reputation, performance metrics, and model similarity:

$$w_i = \beta_{\text{rep}} \cdot \tilde{r}_i + \beta_{\text{metric}} \cdot \tilde{m}_i + \beta_{\text{sim}} \cdot \tilde{s}_i, \quad (27)$$

where $\tilde{r}_i$ is the normalized reputation score: $\tilde{r}_i = \frac{r_i}{\sum_{j=1}^N r_j}$, $\tilde{m}_i$ is

the normalized performance score: $\tilde{m}_i = \frac{1/(1+\mathcal{L}_i^{\text{val}})}{\sum_{j=1}^N 1/(1+\mathcal{L}_j^{\text{val}})}$ and $\tilde{s}_i$ is

the average model similarity: $\tilde{s}_i = \frac{1}{N-1} \sum_{j \neq i} \text{cosine}(\Theta_i, \Theta_j)$. The weight coefficients satisfy $\beta_{\text{rep}} + \beta_{\text{metric}} + \beta_{\text{sim}} = 1$, with typical settings of 0.5, 0.3, 0.2.

Personalized model aggregation strategy. For the target client c_t , top- k similar clients are selected based on data feature similarity:

$$\mathcal{S}_t = \text{topk}(\{\cosine(\mathbf{f}_t, \mathbf{f}_i) \mid i \neq t\}) \cup \{t\}, \quad (28)$$

where $\mathbf{f}_i$ is the data feature vector of the client i . Personalized aggregation weights are computed as:

$$w_{i \rightarrow t} = \frac{w_i \cdot \cosine(\mathbf{f}_t, \mathbf{f}_i) \cdot \gamma_i}{\sum_{j \in \mathcal{S}_t} w_j \cdot \cosine(\mathbf{f}_t, \mathbf{f}_j) \cdot \gamma_j}, \quad (29)$$

where $\gamma_i = 1/(1 + \mathcal{L}_i^{\text{val}})$ is the performance adjustment factor. The aggregated model for the target client is:

$$\Theta_t^{\text{global}} = \sum_{i \in \mathcal{S}_t} w_{i \rightarrow t} \cdot \Theta_i. \quad (30)$$

Adaptive personalized blending. The final client model is obtained through linear interpolation between global aggregation results and local models:

$$\Theta_t^{\text{final}} = (1 - \alpha_t) \cdot \Theta_t^{\text{global}} + \alpha_t \cdot \Theta_t^{\text{local}}. \quad (31)$$

The personalization coefficient α_t is dynamically adjusted based on client reputation:

$$\alpha_t = \begin{cases} \min(\alpha_t + \Delta_\alpha, \alpha_{\max}) & \text{if } \mathcal{L}_t^{\text{val}} < \text{median}(\{\mathcal{L}_i^{\text{val}}\}) \\ \max(\alpha_t - \Delta_\alpha, \alpha_{\min}) & \text{otherwise} \end{cases}, \quad (32)$$

where Δ_α is the adjustment step, and $\alpha_{\max}$ and $\alpha_{\min}$ are the upper and lower bounds for personalization, respectively.

This mechanism ensures that high-performing clients retain more personalized characteristics, while low-performing clients tend to rely more on global knowledge.

Through this blockchain-based reputation FL framework, the system can effectively screen high-quality client contributions while suppressing the influence of low-quality or malicious clients, achieving robust and personalized metro flow prediction model training while preserving data privacy.

2.6 Loss Function Design

The training objective for metro flow prediction is formulated as a multi-task regression problem, where both inflow and outflow predictions are simultaneously optimized. To address the dual challenges of typical flow variations and anomalous passenger surges in metro systems, a carefully designed loss framework is proposed.

Huber loss for robust optimization. The Huber loss is employed as the core component to balance efficient learning during normal conditions with robustness during anomalous events. This hybrid approach provides quadratic convergence for typical flow variations while maintaining linear robustness for sudden passenger surges, preventing the model from being overly sensitive to outliers while ensuring efficient gradient propagation:

$$L_\delta(a, b) = \begin{cases} \frac{1}{2}(a - b)^2 & \text{for } |a - b| \leq \delta \\ \delta(|a - b| - \frac{\delta}{2}) & \text{otherwise} \end{cases}. \quad (33)$$

Composite loss formulation. Building upon the robust foundation of Huber loss, a composite loss function is designed to leverage the complementary strengths of multiple loss components:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{mse}} \cdot \mathcal{L}_{\text{mse}} + \lambda_{\text{mae}} \cdot \mathcal{L}_{\text{mae}} + \lambda_{\text{huber}} \cdot \mathcal{L}_{\text{huber}}, \quad (34)$$

Table 1: The summary statistic of two metro flow datasets.

Datasets	Stations	Connections	Steps	Intervals
Shanghai	288	958	6,716	15 min
Hangzhou	80	248	1,825	15 min

where the Huber loss component is computed as:

$$\mathcal{L}_{\text{huber}} = \frac{1}{2B} \sum_{b=1}^B \left[L_\delta(y_b^{\text{in}}, \hat{y}_b^{\text{in}}) + L_\delta(y_b^{\text{out}}, \hat{y}_b^{\text{out}}) \right]. \quad (35)$$

This multi-component strategy ensures balanced optimization across different operational scenarios in metro systems.

Optimization framework. To ensure stable training across heterogeneous clients in the FL environment, gradient clipping is applied during local training:

$$\text{clip}(\mathbf{g}, c) = \begin{cases} \mathbf{g} & \text{if } \|\mathbf{g}\| \leq c \\ c \cdot \frac{\mathbf{g}}{\|\mathbf{g}\|} & \text{otherwise} \end{cases}. \quad (36)$$

The Adam optimizer with learning rate scheduling provides adaptive optimization:

$$\theta_{t+1} = \theta_t - \eta_t \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}}. \quad (37)$$

This hierarchical loss framework progresses from robust individual components to an integrated optimization strategy, ensuring both efficient convergence during normal conditions and resilient performance during anomalous events in metro systems.

3 Experiments

This section presents a series of experiments to comprehensively evaluate the MF³ model, including comparative analysis, ablation studies, performance analysis, and few-shot experiment. The source code and experimental settings are available at <https://github.com/jixiang10/MF3-code>

3.1 Datasets and Evaluation Metrics

Two real-world metro flow datasets, Shanghai metro and Hangzhou metro [20], covering large networks with numerous stations and extensive historical passenger flow records, are used for evaluation. Both datasets provide detailed spatiotemporal information as directed graphs with stations as nodes and tracks as edges, and time is discretized into 15-minute intervals. Table 1 summarizes key statistics, showing Shanghai Metro is larger with longer sequences, while Hangzhou Metro, though smaller, exhibits rich temporal variation, supporting robust assessment of model generalization across network scales.

In the prediction setting, each dataset is partitioned into training, validation, and testing sets with a ratio of 70%, 10%, and 20%, respectively. Three widely used evaluation metrics are adopted to assess model performance: mean absolute error (MAE), root mean square error (RMSE), and mean absolute percentage error (MAPE). The detailed computation formulas for these metrics are provided in the Appendix C.1.

Table 2: Performance comparison on Shanghai and Hangzhou datasets.

Models	Shanghai											
	15min			30min			45min			60min		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
LSTM	33.76	34.37	60.48	49.16	41.32	58.91	45.42	45.13	68.56	47.25	49.06	70.93
SVR	36.61	42.58	50.69	41.82	46.62	57.38	48.35	48.62	57.44	52.57	58.13	60.48
ReDyNet	21.22	40.58	<u>17.78</u>	21.78	42.17	<u>18.03</u>	22.46	44.09	<u>18.39</u>	23.21	46.28	<u>18.89</u>
PDFormer	22.25	38.58	18.68	22.54	39.43	18.94	23.19	42.49	19.52	23.83	42.92	20.84
FGNN	24.48	46.61	21.23	25.58	48.64	21.91	27.06	51.54	22.65	28.79	55.08	24.07
STDGRL	22.02	42.28	18.73	22.52	43.16	19.23	23.64	45.61	20.12	24.58	47.91	23.52
AGCRN	22.37	45.92	21.17	23.69	48.01	24.29	24.66	49.51	23.26	26.15	49.51	29.16
STNNs	26.87	51.20	21.90	27.56	51.78	25.35	27.37	52.33	23.09	29.38	55.13	33.19
LLGformer	23.47	26.59	23.47	29.44	32.20	28.44	23.19	26.71	23.19	24.60	27.86	24.61
S-DGNN	<u>12.94</u>	<u>15.68</u>	29.28	<u>12.03</u>	<u>14.83</u>	26.25	<u>11.06</u>	<u>13.28</u>	24.62	<u>9.81</u>	<u>12.96</u>	22.23
MF ³	5.81	8.26	10.97	8.75	11.00	14.28	8.29	10.61	14.19	7.13	9.72	12.59
Improvement (%)	55.10	47.32	38.29	27.27	25.83	20.80	25.05	20.11	22.84	27.32	25.00	33.35

Models	Hangzhou											
	15min			30min			45min			60min		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
LSTM	34.02	36.56	60.32	38.31	40.95	68.76	45.23	44.62	69.35	46.73	48.62	70.64
SVR	36.84	42.78	50.23	41.52	47.91	57.23	48.09	48.61	58.19	53.62	59.93	60.77
ReDyNet	21.22	37.51	<u>18.65</u>	22.18	38.28	19.12	22.67	38.93	19.53	23.25	40.00	<u>19.82</u>
PDFormer	22.35	37.45	21.63	21.98	38.83	<u>18.88</u>	22.34	40.78	<u>19.34</u>	23.02	42.80	19.97
FGNN	27.36	48.33	27.19	28.86	51.83	27.49	30.64	56.26	29.10	32.57	60.88	30.74
STDGRL	23.62	47.17	24.97	24.09	51.67	26.09	25.06	51.67	25.06	25.73	51.67	23.03
AGCRN	23.16	41.38	22.78	24.61	45.23	24.32	26.19	49.84	26.09	27.56	53.88	26.58
STNNs	29.93	50.47	26.91	28.45	52.12	25.40	29.36	54.76	25.91	30.12	56.63	26.31
LLGformer	23.68	27.88	23.68	27.08	32.49	27.08	36.07	29.39	26.07	29.74	32.19	29.74
S-DGNN	<u>11.49</u>	<u>14.80</u>	27.14	<u>11.26</u>	<u>13.86</u>	24.29	<u>10.71</u>	<u>13.82</u>	22.40	<u>8.91</u>	<u>12.20</u>	20.83
MF ³	6.29	7.86	17.31	5.22	7.12	15.80	6.61	9.50	18.70	6.29	7.86	17.31
Improvement (%)	45.26	46.89	7.18	53.64	48.63	16.31	38.28	31.26	3.31	29.41	35.57	12.66

3.2 Comparative Analysis

To comprehensively evaluate the performance of the proposed method, a series of comparison experiments are designed. The baselines can be categorized into two groups: the LSTM and the SVR are classical time series prediction methods, serving to verify the basic effectiveness of the model; the subsequent methods are state-of-the-art approaches proposed in recent years, which have demonstrated strong capability in modeling complex spatiotemporal dependencies and dynamic variations in metro or traffic flow prediction.

The comparison results of all baseline methods and the proposed approach are summarized in Table 2, where **bold** and underline denote the optimal and suboptimal values, respectively. As shown, the proposed method consistently outperforms the baselines across multiple evaluation metrics and time intervals. The calculated improvement percentages quantitatively demonstrate its superior effectiveness. Specifically, it achieves a remarkable performance gain over the best baseline by reducing the error metrics by up to 55.10% in MAE, 38.29% in MAPE, and 47.32% in RMSE on the

Shanghai dataset. Similar substantial improvements are observed on the Hangzhou dataset, confirming the model’s robustness and exceptional capability in capturing complex spatiotemporal dependencies for precise metro flow prediction in dynamic real-world scenarios.

3.3 Ablation Studies

To validate the effectiveness of each module in the proposed MF³ framework, ablation experiments were conducted by removing different components and comparing the prediction performance. The results are illustrated in Figure 3. It can be observed that removing the MA module leads to a noticeable degradation across all three metrics, indicating its importance in integrating heterogeneous modal information. Furthermore, eliminating the DMT results in further performance decline, which highlights the necessity of this module for capturing complex spatiotemporal dependencies. In contrast, the removal of the BFR mechanism causes a dramatic deterioration, where MAE, RMSE, and MAPE increase to 18.47, 21.12, and 27.04, respectively, far worse than the complete model. This

demonstrates that BFR plays a crucial role in ensuring robustness and trustworthiness during federated collaboration. Overall, the three modules complement each other and collectively guarantee the superior performance of MF³ in metro flow prediction.

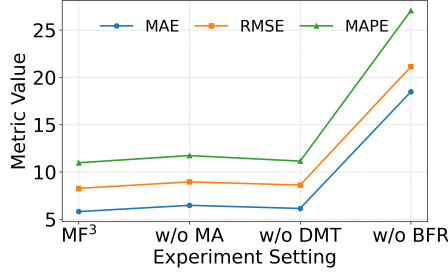


Figure 3: Ablation experiments on the Shanghai dataset.

3.4 Performance Analysis

The prediction performance of MF³ in Shanghai and Hangzhou is analyzed. By examining error trends and client distributions, convergence, accuracy, and fairness are evaluated.

Figure 4 shows that MAE and RMSE decrease steadily with training rounds in both Shanghai and Hangzhou, converging to similarly low values; 15-min and 30-min intervals yield the lowest errors. The parallel trends across cities and intervals confirm robustness and stable outlier handling. Additional results are given in Appendix C.3.

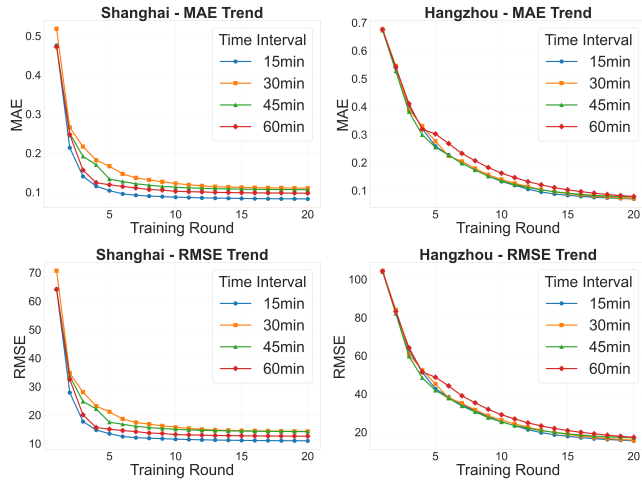


Figure 4: MAE and RMSE of MF³ over training rounds in Shanghai and Hangzhou.

In summary, the experimental results demonstrate that the MF³ achieves good convergence, stability, and fairness across different cities and time intervals, effectively supporting multi-city, multi-modal metro flow prediction tasks.

3.5 Few-Shot Experiment

To evaluate the performance of MF³ under limited training data, a few-shot experiment is conducted. In this setting, the training, validation, and test sets are split in a ratio of 1:2:7, ensuring that effective feature representations can be learned even with scarce training data. The model is trained and tested on data from different time intervals and cities to examine its convergence and prediction accuracy under low-data conditions.

Table 3 presents the few-shot learning performance on Shanghai dataset across different time intervals:

Table 3: The few-shot learning performance on Shanghai dataset.

Intervals	MAE	RMSE	MAPE
15min	14.38	16.65	22.38
30min	19.51	21.68	28.74
45min	20.38	22.67	29.61
60min	10.79	13.66	17.42

Experimental results show that even with 10% data, MF³ outperforms state-of-the-art baselines with significantly lower MAE, demonstrating its exceptional data efficiency.

4 Conclusion

This paper proposes MF³, a privacy-preserving multimodal FL framework for metro flow prediction within WoT-enabled smart city infrastructures. It integrates the MA module for cross-modal fusion, the DMT module for spatiotemporal modeling, and the BFR module for fair aggregation. Experiments on Shanghai and Hangzhou datasets show that MF³ consistently outperforms existing methods in accuracy, stability, and few-shot performance, demonstrating its effectiveness in real-world WoT scenarios.

Future work will incorporate advanced few-shot and zero-shot learning to improve adaptability under limited or unseen data, and leverage transfer learning across diverse cities and time domains to enhance generalization and scalability, making the framework more flexible and intelligent for robust forecasting amidst complex and stochastic passenger behaviors. This will further strengthen the model's practical applicability and deployment potential in dynamic real-world metro systems.

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Appendix

A Related Work

This section summarizes related work on metro flow prediction from three perspectives: multimodal learning for traffic flow prediction, deep learning-based metro flow prediction, and federated learning in traffic flow prediction.

A.1 Multimodal Learning for Traffic Flow Prediction

Metro flow prediction naturally extends to the broader field of traffic flow prediction, where data are inherently multimodal. Each modality, such as temporal, spatial, visual, and contextual, provides unique and complementary information about complex urban dynamics. Compared with unimodal methods, multimodal learning integrates heterogeneous signals, enhancing robustness, interpretability, and generalization across diverse scenarios [23]. Visual modalities demonstrate particular advantages due to their ability to encode high-level spatial correlations and contextual semantics. Unlike traditional tabular data, visual representations preserve structural continuity and latent relationships within traffic environments. For instance, VisionTS [3] converts temporal sequences into image-like matrices by segmenting and stacking them into heatmaps, allowing visual models to perceive periodic patterns. Similarly, VisionSTFT [5] transforms time series into spectrograms using short-time Fourier transform, enabling visual networks to capture frequency-domain dependencies that complement temporal features. In the domain of traffic flow prediction, CNN-based models extract local spatial patterns from grid-based traffic maps, while RNNs and GNNs capture temporal evolution and inter-station dependencies. For instance, DeepST [36] partitions cities into regular grids and employs CNN layers to learn spatial correlations among adjacent regions. DiffusionLSTM [18] combines LSTMs with graph convolutions to model both temporal variations and diffusion-based spatial dependencies. More recent frameworks adopt attention-based fusion strategies to align multimodal representations and emphasize the most informative components [2, 35]. For instance, MMGAT [35] integrates graph attention mechanisms for adaptive multimodal traffic prediction, capturing cross-modal spatiotemporal dependencies efficiently. Multimodal traffic flow prediction remains challenging, as modality alignment is hindered by data discrepancies, while image-based representations increase computational overhead. Consequently, efficient and scalable frameworks capable of fusing temporal, spatial, and visual modalities are still required.

A.2 Deep Learning-Based Metro Flow Prediction

Compared with general traffic flow prediction, metro flow prediction exhibits unique characteristics. Metro routes and operation schedules are largely fixed, and passenger flow patterns show strong periodicity and spatial regularity. These characteristics constrain model flexibility while emphasizing structured dependencies among stations and time intervals. Deep learning significantly advances metro flow prediction by capturing nonlinear dependencies and long-term temporal correlations. RNNs and their variants such as LSTM and GRU model sequential patterns in metro flow data.

For instance, At-STGCN-BiLSTM [19] combines graph convolution, bidirectional LSTM, and attention mechanisms to enhance short-term prediction performance in large-scale metro systems. GNNs capture spatial correlations and dynamic interactions among stations, leveraging the fixed non-Euclidean topology of metro networks. For instance, ReDyNet [8] adapts to variations in metro flow and external environments through a responsive mechanism to construct an appropriate dynamic graph. Transformer-based architectures model long-range dependencies and extract global patterns through attention mechanisms. CSP-AIT-Net [28] applies spatiotemporal graph attention with asynchronous inflow tracking and semantic embeddings for origin–destination flow prediction, demonstrating the effectiveness of attention for metro OD prediction. Emerging architectures such as Mamba provide linear computational complexity and strong local feature extraction capability for long sequence modeling. For instance, ST-MambaSync [24] embeds Mamba blocks into a spatial–temporal synchronization network, capturing lane-level speed surges within long multi-sensor sequences and enabling minute-by-minute traffic prediction for urban expressways. Despite these advances, metro-specific challenges remain. Fixed routes and timetables impose rigid spatial-temporal patterns, amplifying the impact of peak-hour fluctuations and service disruptions. The strong periodicity and dense station topology increase the difficulty of generalizing across days or lines. Furthermore, real-time metro operation requires models that balance global dependency modeling and computational efficiency. Achieving such a balance between prediction accuracy, scalability, and responsiveness continues to be an open research problem.

A.3 Federated Learning in Traffic Flow Prediction

Traffic flow prediction often faces challenges related to privacy, data heterogeneity, and decentralization. Raw data cannot always be shared across regions or cities due to privacy concerns, and centralized modeling may lead to biased or insecure predictions. FL addresses these issues by allowing local model training at each station or client while aggregating updates at a central server, thus mitigating data silos and privacy leakage [38]. Prior studies demonstrate the effectiveness of FL in traffic flow prediction while preserving privacy. For instance, FedTPCS [39] introduces a two-phase client selection strategy to minimize energy consumption and latency by addressing device and data heterogeneity. FUELS [21] employs dual semantic alignment-based contrastive learning to capture spatio-temporal heterogeneity. Additionally, 2MGTCN [33] combines graph convolutional networks and temporal convolutional networks with federated transfer learning to enhance cross-city prediction accuracy and privacy preservation. Nevertheless, designing FL frameworks that handle station heterogeneity, real-time metro flow fluctuations, and privacy-preserving multi-station collaboration remains challenging.

Despite significant advances in multimodal learning, deep learning-based metro flow prediction, and federated learning, several overarching challenges remain. Efficiently integrating heterogeneous modalities while preserving spatial-temporal dependencies, capturing both global patterns and local dynamics in highly periodic and dynamic metro networks, and ensuring privacy-preserving,

Algorithm 1 The MF³ framework for metro flow prediction**Require:** Historical metro data $\mathbf{X}$, spatial graph $\mathbf{A}$, heatmaps $\mathbf{H}$ **Ensure:** Predicted passenger flows $\hat{\mathbf{Y}}$

```

1: Initialize: Global model  $\Theta_G$ , client reputations  $\mathcal{R}$ 
2: for round = 1 to  $R$  do
3:   // Client Local Training
4:   for each client  $c_i$  in parallel do
5:     Load local data  $\mathcal{D}_i$ 
6:     Initialize local model  $\Theta_i \leftarrow \Theta_G$ 
7:     for epoch = 1 to  $E$  do
8:       Process data through the MA and the DMT modules
9:       Compute composite loss  $\mathcal{L}_{\text{total}}$ 
10:      Update  $\Theta_i$  with gradient clipping
11:    end for
12:    Compute validation loss  $\mathcal{L}_i^{\text{val}}$ 
13:    Upload  $\langle \Theta_i, \mathcal{L}_i^{\text{val}} \rangle$  to server
14:  end for
15:  // Server Aggregation
16:  for each client  $i$  do
17:    Update reputation:  $r_i \leftarrow \alpha \cdot r_i + (1 - \alpha) \cdot \phi(\mathcal{L}_i^{\text{val}})$ 
18:  end for
19:  Compute aggregation weights based on reputation and
  performance
20:  Identify similar clients for each target client
21:  for each target client  $t$  do
22:    Compute personalized weights  $w_{i \rightarrow t}$ 
23:    Aggregate:  $\Theta_t^{\text{global}} \leftarrow \sum w_{i \rightarrow t} \cdot \Theta_i$ 
24:    Adjust personalization coefficient  $\alpha_t$ 
25:    Blend:  $\Theta_t^{\text{final}} \leftarrow (1 - \alpha_t) \Theta_t^{\text{global}} + \alpha_t \Theta_t^{\text{local}}$ 
26:  end for
27:  Update global model  $\Theta_G$ 
28:  Distribute updated models to clients
29: end for
30: // Prediction Phase
31: function PREDICT(historical data, current time  $T$ )
32:   Preprocess: Normalize flows, extract features, generate
   heatmaps
33:   Multimodal Alignment: Fuse temporal, spatial, visual data
34:   DMT Processing:
35:     Temporal path: Fine-grained + periodic patterns
36:     Spatial path: Convolutional feature extraction
37:     Cross-attention fusion between pathways
38:   Generate predictions  $\hat{\mathbf{Y}}$  for  $T + 1$  to  $T + p$ 
39:   return  $\hat{\mathbf{Y}}$ 
40: end function
41: Output: Trained models for all clients, prediction function

```

fair, and scalable collaborative learning across decentralized stations continue to pose significant difficulties for practical metro flow prediction.

B Algorithm Pseudocode and Complexity Analysis

The MF³ algorithm framework integrates multimodal data processing with privacy-preserving FL for metro flow prediction. As shown

in Algorithm 1, the framework operates through iterative client-server interactions while maintaining data privacy through local model training.

Time complexity analysis. Let N be the number of clients, E the local epochs, B the batch size, T the sequence length, and S the number of stations. The computational complexity is primarily determined by four components: the MA module requires $O(T \cdot S \cdot d^2)$ operations for cross-modal attention mechanisms; the DMT module exhibits $O(T \cdot S \cdot d + T \cdot d^2)$ complexity by combining the linear scaling of Mamba with attention mechanisms; client training dominates the overall cost with $O(N \times E \times (|D_i|/B) \times (T \cdot S \cdot d^2))$ where d represents the model dimension and $|D_i|$ denotes client i 's data volume; and the BFR module contributes $O(N^2 \cdot |\Theta|)$ for similarity computations and model averaging during server aggregation. The framework's design ensures scalability for large-scale metro networks while maintaining the privacy guarantees of FL through efficient distributed computation.

C Experiment

This section provides supplementary details for the experiments presented in the main text.

C.1 Evaluation Metrics

To quantitatively assess the prediction performance of different models, three commonly used evaluation metrics are employed: MAE, RMSE, and MAPE. Their mathematical formulations are provided below:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|, \quad (38)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}, \quad (39)$$

$$\text{MAPE} = \frac{100\%}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i + \epsilon} \right|, \quad (40)$$

where y_i and $\hat{y}_i$ represent the ground truth and predicted values, respectively, N denotes the total number of samples, and ϵ is a small constant to prevent division by zero. Note that for all three metrics, lower values indicate better prediction performance, as they reflect smaller deviations between the predicted and true values.

C.2 Baseline Methods

To ensure a comprehensive and fair evaluation, the baseline models are grouped into classical methods, graph-based models, and transformer-based or advanced spatiotemporal deep learning methods:

Classical methods:

LSTM [12]: A classical recurrent neural network capable of capturing long-term dependencies in historical flow data, widely used in traffic and metro flow prediction. LSTM serves as a baseline to evaluate the benefits of more advanced graph or transformer-based architectures over standard sequence modeling.

SVR [6]: A traditional regression method suitable for small-sample or nonlinear time series prediction. SVR represents classical

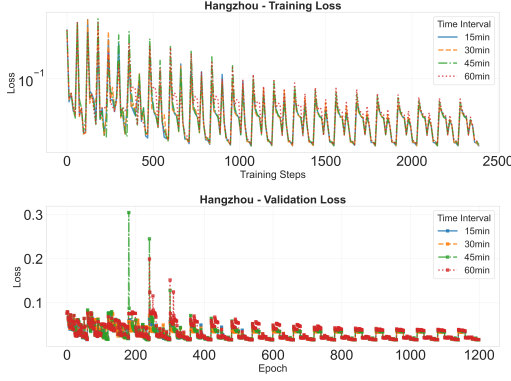
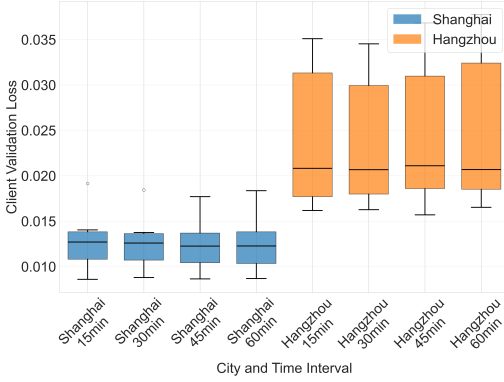


Figure 5: Hangzhou metro loss curves.

Figure 6: The client performance under MF^3 in FL.

statistical approaches, providing a baseline for assessing the value added by deep learning and multimodal fusion techniques.

Graph-based models:

ReDyNet [8]: A responsive dynamic graph neural network designed for metro flow. ReDyNet constructs dynamic graphs and disentangles redundant signals to achieve accurate spatiotemporal modeling, representing state-of-the-art graph-based approaches for metro networks with complex station interactions.

FourierGNN [32]: Reformulates multivariate time series as hypervariate graphs and applies graph convolutions in Fourier space to efficiently capture spatiotemporal dynamics. FourierGNN exemplifies spectral-domain GNN methods for high-dimensional traffic data modeling.

STDGRL [29]: Utilizes spatiotemporal node embeddings and dynamic graph relationship learning combined with a transformer module for long-term flow prediction, highlighting the integration of graph learning with sequential modeling.

AGCRN [1]: Incorporates node-adaptive and data-adaptive modules to automatically capture fine-grained spatiotemporal correlations without requiring a predefined graph, representing flexible GNNs for heterogeneous urban traffic data.

S-DGNN [4]: Models transfer stations as graph nodes, leveraging spatiotemporal correlations with GRU to predict short-term flow, aiding operational decisions.

Transformer-based and advanced spatiotemporal models:

PDFormer [16]: A propagation delay-aware long-range transformer that captures short- and long-range spatial dependencies while modeling the time delay of information propagation, illustrating the advantage of transformers in handling long-range spatiotemporal patterns.

STNNs [30]: Employs spatial and temporal transformers with multi-head attention to jointly model directed spatial dependencies and long-range temporal dependencies, providing a representative baseline for scalable long-term prediction.

LLGformer [17]: Constructs a learnable long-range spatiotemporal graph and applies attention mechanisms to capture complex patterns across time and space, improving traffic and metro flow prediction efficiency.

C.3 Performance Analysis

Figure 5 shows the training and validation loss curves over training rounds for different time intervals in Hangzhou.

It can be seen that the model continuously converges on the training set, and the 60-minute interval exhibits more stable validation performance, indicating that coarser-grained predictions achieve more reliable generalization.

Figure 6 presents the client performance distribution using box-plots. In the figure, the box represents the interquartile range (middle 50% of clients), the line inside the box denotes the median, the whiskers indicate the normal range of the data, and the outliers reflect clients with exceptional performance.

It can be observed that client performance is generally stable across different configurations, with medians concentrated, indicating good fairness of MF^3 in FL and limited performance variation among clients.

C.4 Statistical Significance Analysis

To verify reliability, paired t-tests are conducted across five independent runs with distinct seeds. As visualized in Fig. 7, MF^3 consistently achieves the lowest errors. Significance tests against both the second-best baseline and ablation variants confirm statistically significant improvements ($p < 0.05$), validating the contribution of individual modules and the framework's overall robustness. Furthermore, the narrower error bars observed for MF^3 indicate superior stability against initialization noise compared to the fluctuating performance of competing methods.

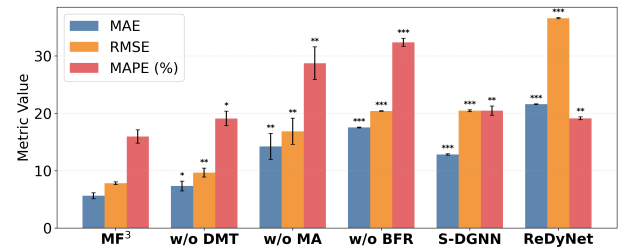


Figure 7: Performance comparison with statistical significance. Error bars: standard deviation. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.